

Blind Multi-Stage Scoring Auctions with Two-Sided Incomplete Information for Government Procurement

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Abstract. Government procurement often suffers from subjectivity because reviewers rely on variable and partially unknown value functions to score multi-attribute bids. We address this challenge with a few-shot value function estimation framework and propose a greedy online heuristic solver that incrementally approximates both the structure and weights of the underlying value function. We apply the algorithm to a public works case study, where our results show that a 7-attribute value function can be accurately recovered in under 20 iterations, enabling bidders to adapt quickly and effectively in multi-stage scoring auctions.

Keywords: government procurement, scoring auctions, few-shot learning, value function estimation, online optimization

1 Introduction

In this paper we analyze the multi-round *best-value* auction style frequently used in public works bidding along with the few shot value function estimation problem that the bidder’s must solve during the course of the auction.

Under this format a panel of “reviewers” will receive n bids from n bidders. This panel then assigns scores across multiple criteria and combines them to form a total score which is returned to the bidder at the end of a round [3]. Such criteria can often times be difficult to quantify. For example, an agency may conduct a *best-value* auction where bidders submit proposals evaluated on technical merit and past performance. In other words the panel has a multi-attribute value function $V(x) : \mathbb{R}^d \rightarrow [0, 100]$ which scores a d attribute bid x and returns a normalized score between 0 and 100. This process continues iteratively with bidders submitting bids and receiving scores during each round until a terminal round where the bidder with the highest score is awarded the government contract.

We argue that this reliance on qualitative evaluation introduces significant uncertainty into the bidding process. Additionally that uncertainty is shaped less by economic fundamentals and more by the personal preferences of the auctioneer. Biases within this context have been well documented and effect both the bidder and the auctioneer. For example, biases like Lower-Bid Bias where a

government agency will tend towards awarding a contract to the lowest bid despite the risk of reduced quality, are commonplace according to an experimental study conducted by Dekel and Schurr. [4].

Due to panel biases and parameters in the panel’s value function that are unknown to the bidder, a bidder is forced to roughly estimate the value function $V(x)$ within a few rounds of bidding in order to efficiently apply inter-round adjustments to their bids. We refer to that problem as “few-shot value function estimation”.

We recognize that the problem of reconstructing a subject’s preferences through queries is well studied. However literature focused on preference elicitation grants the “questioner” and subject far more flexibility in what type of questions and answers are asked and given. For example the “questioner” may query the subject on local attributes to elicit marginal values/utilities or even ask for the subject’s cardinal ranking of attributes [1][2]. In contrast, our approach addresses a more restrictive yet realistic case, where user’s only receive an aggregated score, with no decomposable feedback available. To our knowledge this specific auction-style setting of multi-attribute value function estimation through global aggregated scoring has not been addressed in existing literature.

And so we proceed by first focusing our attention on developing a general algorithm to estimate a value function with only a few (input,output) pairs. Next we elicit a value function from an expert in the field of public works construction bidding. Finally we simulate a multi-round bidding process, where a bidder, whose goal is to produce a bid with maximum value, uses our proposed algorithm to quickly estimate the elicited value function within a few rounds of simulated bidding.

2 Few Shot Value Function Estimation

To accomplish few-shot value function estimation, we first establish a few definitions in order to formalize the problem. First we assume that the true value function $V : \mathbb{R}^d \rightarrow [0, 100]$ is additive over d single attribute value functions $v_i : \mathbb{R} \rightarrow [0, 100]$, each of which is weighted based on importance. As such the value function is designed to attribute a value to a d dimensional input x .

$$V(x) = \sum_{i=1}^d w_i \cdot v_i(x_i), \quad x \in \mathbb{R}^d, w \in \Delta_d$$

$$\Delta_d = \left\{ \sum_{i=1}^d w_i = 1, w \geq 0 \right\}$$

At this stage, its reasonable to assume two features about our value function. First that x is in a bounded and closed subspace of $\mathbb{R}^d$. This is an intuitive simplifying assumption since in most practical cases an input x with infinite value represents an agent with infinite resources which is practically unreasonable. We can simplify our feasible region further by adopting the norm that each

attribute is scored on a range from 0 to 1. If the value function is categorical with n ordered categories like $x_i = \{\text{Good}, \text{Medium}, \text{Bad}\}$ then we assume that the normalized numerical encoding of the single attribute value function domain is $\{\frac{i}{n-1} : i = 1, \dots, n-1\}$. For example $\text{Good} = 1, \text{Medium} = 0.5, \text{Bad} = 0$. From here on out we assume that $x \in [0, 1]^d$. Next we make the simplifying assumption that $v_i(x_i)$ falls within a finite set of parametric functions. For example $v_i(y)$ may be exponential, linear, or triangular. For example an exponential value function would be $v_i(y) = \alpha_i^0 e^{\alpha_i^1 y}$. We denote these prototype single attribute value function forms as $\phi_k(y; \alpha_k) : [0, 1] \times \mathbb{R}^{|\alpha_k|} \rightarrow [0, 100]$ where α_k is a vector of parameters necessary to define the function $\phi_k(x_i)$. We also define a "library" of preselected prototype functions of size M as $\mathcal{F} = \{\phi_k(y; \alpha_k)\}_{k=1}^M$. Now that we've defined the value function V and "prototype" library $\mathcal{F}$ we can define an approximation of V .

$$\tilde{V}(x) = \sum_{i=1}^d w_i \sum_{k=1}^M z_k^i \phi_k(x_i; \alpha_k^i) \quad (1)$$

$$\text{s.t. } z_k^i \in \{0, 1\}, \quad \sum_{k=1}^M z_k^i = 1 \quad \forall i \in \{1, \dots, d\}, \quad w \in \Delta_d \quad (2)$$

In other words we've defined an approximation which must pick one basis function out of a library of size M to serve as the single attribute value function for each of the d dimensions of x , while still ensuring that the weights w are unit. We now transition to the question of finding the best approximation given only a few data points. Given a history of inputs and returned values $\mathcal{H}^t = \{(x^s, V(x^s)) : s = 1, \dots, t\}$ where t is small, we would like to find parameters $\{z_k^i, \alpha_k^i, w_i : k = 1, \dots, M, i = 1, \dots, d\}$ such that our value approximation $\tilde{V}(x)$ is close to the true value function $V(x)$. To this end we define the following mixed integer program, which may be non-linear depending on the library of prototypes $\mathcal{F}$ that we choose.

$$\min_{\{z_k^i, \alpha_k^i\}_{i,k=1}^{d,M}, \{w_i\}_{i=1}^d} \left[\frac{1}{t} \sum_{s=1}^t (V(x^s) - \tilde{V}(x^s))^2 \right] \quad (3)$$

$$\text{s.t. } z_k^i \in \{0, 1\}, \quad \sum_{k=1}^M z_k^i = 1 \quad \forall i \in \{1, \dots, d\}, \quad w \in \Delta_d \quad (4)$$

With the problem of few-shot value function estimation clearly formalized, we can begin exploring a solution method.

2.1 Online Heuristic Greedy Solver

The minimization problem detailed in equation 4 is a combinatorial optimization program that, depending on the basis functions chosen, may be non-linear and

non-convex. As a result the problem is computationally expensive to solve. So we investigate a heuristic greedy solver that splits the problem at each iteration into $d \times M$ subproblems and a master problem.

Assume that we are given a history of $\mathcal{H}^t = \{(x^s, V(x^s)) : s = 1, \dots, t\}$. We begin by fixing $w_i = \frac{1}{d}$ for all $i = 1, \dots, d$, then solve the following optimization problem for each dimension d . This problem represents a greedy solution since it aims to choose basis functions that can best explain the entirety of the observed values for each input dimension.

$$\phi_{k_i}^*(y; \alpha_k^{i*}) = \arg \min_{\phi_k \in \mathcal{F}} \min_{\alpha_k^i} \sum_{(x^s, V(x^s)) \in \mathcal{H}^t} (V(x^s) - w_i \phi_k(x_i^s; \alpha_k^i))^2 \quad (5)$$

Once we've obtained a set of optimal basis functions and their respective optimal parameters for each input dimension, $\{\phi_{k_i}^*(y; \alpha_k^{i*}) : i = 1, \dots, d\}$, we can then solve the master problem which retrieves the optimal weights. Namely we solve

$$w^* = \arg \min_{w \in \Delta_d} \sum_{(x^s, V(x^s)) \in \mathcal{H}^t} \left(V(x^s) - \sum_{i=1}^d w_i \phi_{k_i}^*(x_i^s; \alpha_k^{i*}) \right)^2 \quad (6)$$

Once new data is introduced and we obtain an expanded/richer $\mathcal{H}^{t+1}$ we recompute this process, except instead of initializing $w_i = \frac{1}{d}$ while we solve the next d sub-problems, we allow w to equal w^* from the last iteration. Given that at each iteration we are given new data $(x^{t+1}, V(x^{t+1}))$ the full greedy solution algorithm can be described compactly as follows.

Algorithm 1 Online Greedy Solver

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1: input:  $T$ —Number of Iterations
2: input:  $\mathcal{F}$ —Library of Prototypes
3: input:  $V(x) : [0, 1]^d \rightarrow [0, 100]$ 
4: set:  $w^1 = [\frac{1}{d}, \dots, \frac{1}{d}] \in \mathbb{R}^d$ 
5: set:  $\mathcal{H}^1 = \{(x^1, V(x^1))\}$ 
6: for  $t = 1$  to  $T$  do
7:   for  $i = 1$  to  $d$  do
8:     for  $\phi_k \in \mathcal{F}$  do
9:        $\ell_i[\phi_k(y; \alpha_k^{i*})] \leftarrow \min_{\alpha_k^i} \sum_{(x^s, V(x^s)) \in \mathcal{H}^t} \left( V(x^s) - w_i^t \phi_k(x_i^s; \alpha_k^i) \right)^2$ 
10:    end for
11:     $\phi_{k_i}^*(y; \alpha_k^{i*}) \leftarrow \arg \min_{\phi_k \in \mathcal{F}} \ell_i[\phi_k(y; \alpha_k^{i*})]$ 
12:  end for
13:   $w^* = \arg \min_{w \in \Delta_d} \sum_{(x^s, V(x^s)) \in \mathcal{H}^t} \left( V(x^s) - \sum_{i=1}^d w_i \phi_{k_i}^*(x_i^s; \alpha_k^{i*}) \right)^2$ 
14:   $w^{t+1} \leftarrow w^*$ 
15:   $\mathcal{H}^{t+1} \leftarrow \mathcal{H}^t \cup \{(x^{t+1}, V(x^{t+1}))\}$ 
16: end for

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3 Case Study: Public Works Procurement

Our goal will be to apply the online greedy solver to simulate several rounds of bidding within a public works procurement procedure. Namely a government agency will set up a multi-round best value auction, where bidders can bid to win a construction contract. Within this auction format the bidders submit bids and receive a score corresponding with their bid during every round of bidding. Throughout the rounds each bidder can adjust their bid based on the score received in order to improve their chance of winning the contract. In order to simulate such an auction we must first elicit a value function from an expert in the field of public works procurement. This value function will represent the government’s scoring panel. Then we can apply our greedy online solver to simulate a bidders bidding behavior. To apply our model to the specific case of public works bidding, we first elicit a value function from an expert in the field through the SMART method [5]. We chose to interview a construction manager whose been working in the industry for approximately 40 years [6]. We first interviewed the expert and elicited an objective hierarchy. Next we applied the swing weight procedure to determine the importance ranking and the weights of each objective within the value function. Both the elicited attributes and swing weights are detailed in table 1. Swing weights are usually considered to be the most accurate method for deriving attribute weights [7].

Attribute	Swing Rank	Swing Rating	Swing Weight
Cost	1	100	0.1812
Number of Similar Projects	2	90	0.1630
Personnel Experience	3	87	0.1576
Safety Score	4	85	0.1540
Projected Duration	5	70	0.1268
Project Manager Procedure	6	65	0.1178
On-Time Completion History	7	55	0.0996

Table 1: Swing Weighting for Construction Bid Attributes

We elicit the value function itself by applying the bisection method for all natural attributes and ranking for constructed attributes. We outline those finding in table 2. We note that cost has a triangular value function which is linear between $[0, 0.9]$ and $[1, 1.25]$. This structure is intuitive since, if a bidder has an expected cost of 0 a reviewer will naturally be skeptical of such a bid and assign a value of 0. On the other hand, a bid that has an expected cost 25% greater than the government’s explicit budget will also be assigned a value of 0 since such a cost would be a disqualifying feature of the bid. We list of critical points of each attribute in figure 2.

Attribute	Point 1 (Value)	Point 2 (Value)	Point 3 (Value)	Point 4 (Value)
Safety Score	≤ 1.0 (100)	2.0 (50)	3.0 (25)	≥ 5.0 (0)
Personnel Experience	Excellent (100)	Acceptable (50)	Below Standard (10)	Unacceptable (0)
# of Similar Projects	0 (0)	3 (30)	7 (70)	≥ 10 (100)
On-Time completion history	0% over (100)	25% over (75)	50% over (50)	100% over (0)
PM Procedure	Excellent (100)	Acceptable (60)	Below Standard (20)	Unacceptable (0)
Projected Duration	20% under (100)	10% under (75)	On target (50)	20% over (0)
Cost	$0 \times$ (0)	$.9 \times$ (100)	$1 \times$ (90)	$\geq 1.25 \times$ (0)

Table 2: Representative Value Function Points for Each Attribute

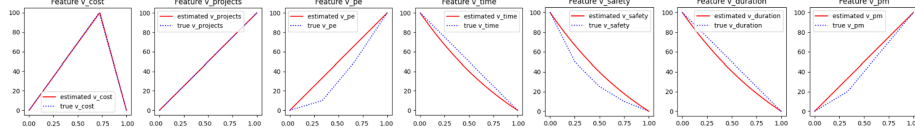
In order to apply the greedy online solver described in 1 to simulate an auction we first define a library of single-attribute value functions $\phi_k(x) : [0, 1] \rightarrow [0, 100]$. We choose the following library $\mathcal{F}$, which captures positive and negative linear/exponential and "tent"-shaped functions.

$$\mathcal{F} = \{\phi_1, \phi_2, \phi_3, \phi_4, \phi_5\} \quad (7)$$

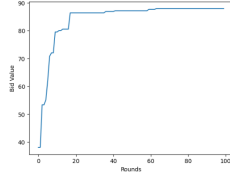
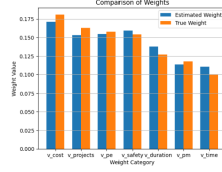
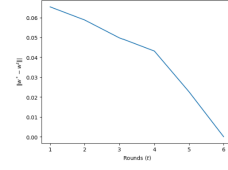
$$\phi_1(x; \alpha_1) = 100x, \quad \phi_2(x; \alpha_2) = -100x + 100, \quad \phi_3(x; \alpha_3) = \frac{e^{\alpha_3 x} - 1}{e^{\alpha_3} - 1} \quad (8)$$

$$\phi_4(x; \alpha_4) = 100 \cdot \frac{2 - (e^{\alpha_4 x})}{e^{\alpha_4} - 1}, \quad \phi_5(x; \alpha_5) = \begin{cases} \frac{100x}{\alpha_5}, & \text{if } x \leq \alpha_5 \\ 100 - \frac{100(x - \alpha_5)}{1 - \alpha_5}, & x > \alpha_5 \end{cases} \quad (9)$$

We must also formally define the value function elicited. Note that there are 7 objectives outlined in table 1 and therefore we have 7 corresponding single attribute value functions which we denote to be v_i . v_i is defined to be the linear interpolation of the points in table 2, however as discussed above we normalize all inputs to be in $[0, 1]$. To begin our simulation we allow the fictitious bidder to begin with a random bid $x^0 \in \text{Uniform}([0, 1]^d)$. Then at the beginning of every round the bidder will maximize their current approximation of the auctioneer's value function $\tilde{V}(x)$ and choose their next bid as $x^{t+1} \leftarrow \arg \max_{x \in [0, 1]^d} \tilde{V}(x)$. This is inline with a competitive bidding environment where T is unknown to the bidders and so each bidder must submit their best bid at every round in order to hedge against the current round being their last. In figures 1,3 we display the results of how well each single attribute value function was estimated, along with the estimated weight of each single attribute objective. We find that at the 19th iteration the correlation between $\tilde{V}(x)$ and $V(x)$ is approximately $\rho = 0.989$, when averaged over 10,000 uniformly random choices of x^0 .

Fig. 1: Approximated single attribute value functions after $T = 20$ rounds

In figure 2 we provide the score achieved at every round of the bidding simulation. The online greedy value function estimator is able to find a high value bid along with a rough estimate of the auctioneer's value function structure, within a few rounds. This would imply that the online greedy solver is able to find high value bids faster than it can find the structure of the value function. However because of the problem's non-convexity, the algorithm has a tendency to plateau for several rounds. Meaning that a bidder would struggle to pin down the exact value function structure and as a result would be incapable of producing a bid of higher value than its last submission.

Fig. 2: Value of Best Bid at round $t = 1, \dots, 100$ Fig. 3: Approximated weights for each objective after $T = 20$ Fig. 4: Approximated weights for each objective after t rounds, where sub-value functions $\{\phi_i\}_{i=1}^d$ are known

We also observe that most multi-bid auctions will likely last 3-5 rounds, which is too few rounds for algorithm 1 to yield an accurate estimation. To that end we note that if the bidder is familiar with the structure of the auctioneer's value function, then the problem reduces to finding the weights associated with each attribute by solving equation 10, where $\mathcal{H}^t$ is the set of all (bid, score) pairs collected by the t^{th} round. In figure 4 we show that even after 1 round of bidding $\|w^* - w\|_2^2 = .07$ and after 6 rounds $w^* = w$, where w^* are the true weights and w are the estimated weights.

$$\min_{w \in \Delta_d} \frac{1}{2} w^T Z^T Z w - w^T Z v, \quad Z = [\phi_{k_i^*}(x_i^s; \alpha_k^{i*})]_{s,i=1}^{t,d} \in \mathbb{R}^{t \times d}, \quad v = [V(x^s)]_{s=1}^t \quad (10)$$

4 Conclusion

Throughout this paper, we explored the problem of blind multi-stage scoring auctions with two-sided incomplete information. We identified the problem of few-shot value function estimation as key step in the bidding process of such an auction. To that end we introduced a greedy online solver that approximates the structure of an unknown multi-attribute value function using a limited number of observed bids and their associated scores. Our algorithm breaks the approximation problem into tractable sub-problems then returns the shape of each objective’s value function along with their corresponding weights.

We applied the proposed method to a case study in public works construction bidding. We first elicited a realistic value function through an expert interview, then simulated a bidder’s bidding behavior using online value function estimation. Our results indicate that the greedy online solver can quickly identify high-value bids and accurately approximate both the shape and weights of the government’s value function within a small number of rounds. However, we also identify certain limitations. Namely that the algorithm tends to plateau due to local non-convexity. Further work may include exploring the efficacy of hybrid methods that combine greedy search with neural network-based meta-learning strategies. Overall, our results suggest that few-shot learning algorithms can substantially mitigate the strategic uncertainty inherent in multi-stage scoring auctions.

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